

uploaded uPs By Project Name James Brakefield

year prefix	name	aliases	type	idea date	bus. case	formal doc	abrev docx	xlxs	RTL	inst sz	data sz	register sz	address sz	adr modes	data types	project name	FPGA part	FPGA tool	run date	LUTs	DFFs	DSP	Block rams	clks/ inst	ns	wd sz adj	std FOM	TB	TB OK	FPGA bd	github locations	comments	
1999	BCD1	ALT1620, radix10	BCD	06/01/92			Y	Y		v4	v4																					variable length BCD arithmetic, coprocessor	
2000	quadISA	alta	cisc	05/03/00	Y		Y	Y								quadISA																all_projects/2000_quadISA	hardware macro facility, no detailed ISA xlsx
2002	LEM1_9		LEM	04/21/02	Y	Y	Y	Y	Y	1	1					LEM1_9	S-7	v25.1	08/14/25	79	17		0.5	1	9.5	0.04	53.3	Y	Y		jimbrake/lem1_9min	cond branch, subroutines	
2002	LEM1_9min		LEM	04/21/02	Y	Y	Y	Y	Y	1	1					LEM1_9min	S-7	v25.1	08/14/25	21	36		0.5	1	4.7	0.02	202.6	Y	Y	Y	jimbrake/Hello-Ljorld	no cond branch, no subroutines	
2002	LEM1_9ptr		LEM	04/21/02	Y	Y	Y	Y	Y	1	1					LEM1_9ptr	S-7	v25.1	08/14/25	114	17		0.5	1	9.7	0.04	36.2	Y	Y		jimbrake/lem1_9min	cond branch, subroutines, mem pointers	
2002	LEM4_9		LEM	04/21/02	Y	Y	Y	Y	Y	4	4					LEM4_9	S-7	v25.1	08/14/25	124	21		0.5	1	9.4	0.15	128.7	Y	Y		jimbrake/lem1_9min	cond branch, subroutines	
2002	LEM4_9ptr		LEM	04/21/02	Y	Y	Y	Y	Y	4	4					LEM4_9ptr	S-7	v25.1	08/14/25	172	21		0.5	1	11.5	0.15	75.8	Y	Y		jimbrake/lem1_9min	cond branch, subroutines, mem pointers	
2009	ALT1604	acc18_18	accum	06/06/09			Y									acc18_18																	
2009	LEM8_9		LEM	06/04/09					Y	9	8	8	11			lem8_9	K-7	ISE	06/04/09	92	61		1	1	2.2	0.33	1630	Y	Y			non-pipeline version	
2009	LEM8_9		LEM	06/04/09					Y	9	8	8	11			lem8_9	V-5	ISE	06/04/09	114	89		1	1	4.5	0.33	643.3	Y	Y			pipeline version	
2013	hay4stk		stack	07/23/13			Y	Y	Y							hay4stk16_32	K-7	ISE	06/07/15													needs review as to implementation status	
2014	BCD2	BCD_two, radix100	BCD	09/18/99			Y				v8																					variable length radix100 arithmetic, coprocessor	
2014	LEM16_18		LEM	06/06/09			Y	Y	Y	18	16	16				lem16_18m	S-3	ISE	05/01/13	1214	297		20	2	11.9	0.67	46.6				jcp_uP/LEM/lem16_18M	RISC + bit field ops	
2016	ALT430		maoc	03/05/00	Y		Y	Y	Y																							all_projects/2016_ALT_430	
2016	ROIS24_24		RISC	06/08/16	Y	Y	Y	Y	Y	24	24	24	24			ROIS24_min	S-7	V25.1	08/14/25	402	96	1	0.5	1	9.7	0.8	205.2	Y	Y				
2016	ROIS24_3sz		RISC	07/17/23	Y	Y	Y	Y	Y	24	3a	24		u2f																			
2018	12X_uP		stack	09/11/15			Y	Y	Y	12	12	12	12			the12X_12uP	k-7	ISE	10/24/18	201	31	0	0	1	4.5	0.5						early OOSM, probably not be complete?	
2018	ALT780		maoc	03/05/00	Y		Y	Y	Y																							all_projects/2018_ALT_780	
2018	ALTx86		maoc	03/05/00	Y		Y	Y	Y																							all_projects/2018_ALT_x86	
2020	OOSM	astoc, stoc11, 1osm	stack	12/27/20			Y	Y																								jimbrake/ASTOC11	
2020	ROC32_8		RISC	05/10/21			Y	Y																								jimbrake/ROC32_8	name changed to TROC, no 16-bit instructions
2021	ALT11	ASTOC11, STOC11	maoc	06/28/21			Y																									OOSM & ALT-11 co-developed, later diverged	
2023	1OSM16_3sz	OOSM	stack	11/04/12			Y																										
2023	1OSM24_3sz	OOSM	stack	11/04/12			Y	Y																									RTL probably incomplete
2024	TROC16 basic		TROC	11/08/24	Y	Y	Y	Y	Y	16	16	20	16	u2		troc16_hwa	S-7	V25.2	12/03/25	397	73	1	0.5	1	18.0	0.67	93.8	Y			all_projects/2024_TROC16, jimbrake/TH	TB runs, not verified	
2024	TROC16 full		TROC	12/09/24	Y	Y	Y	Y	Y	16	16	20	16	u2		troc16_hwa	S-7	V25.2	12/03/25	632	72	1	0.5	1	17.8	0.67	59.6	Y			all_projects/2024_TROC16, jimbrake/TH	TB runs, not verified	
2024	TROC24		TROC	12/09/24	Y		Y	Y		3A	3a	32	24	u2fl																		RTL incomplete	
2025	BROOCC		stack	09/03/25			Y																										
26		28		14		6	25	20	14	15		12		11		10	8	1	Column sums														

instruction size codes

9-18: 2A, 8-24-32: 3B

data size codes

8-16-24: 3a

data type U: unsigned, 2's complement, 1's complement, sm: sign-magnitude, bf: basic float, tf: tapered float, T: takum, fln: fixed point log

addressing modes

I: immediate, vl: variable length immediate, r: relative, a: absolute, bi: base+offset, br: base+register, bx: base+index

Column info

year prefix	have prefixed the name with the year of first recorded log file info
name	ISA name
aliases	other ISA names, changes are made over time for better pronunciation or effect
type	BCD : binary coded decimal, RISC : aka ROC register oriented computer, MAOC : aka CISC , stack : both return and data stacks, TROC : tagged ROC, OOSM : one operand stack machine, LEM : logic emulation machine, accum : accumulator dsgn
idea date	per daily log file or written notes
bus. case	business case, why there is a market for this ISA
formal doc	precise instruction by instruction description of the ISA, e.g. programmer's reference manual
abrev docx	basic description without the precision
xlxs	spreadsheet of the instructions and formats, may have opcode binary codes
RTL	VHDL or verilog implementation files
inst sz	code for instruction bit sizes
data sz	code for data bit sizes
register sz	register bit size not including tag bits
address sz	address size in bits
adr modes	code for addressing modes
data types	code for data types
project name	particular development project name
FPGA part	FPGA family
FPGA tool	FPGA development tool and version
run date	date the FPGA run was made
LUTs	LUT count for the FPGA run
DFFs	D flip-flop count
DSP	multiply/accumulator count
Block rams	block RAM count
clks/ inst	average number of clock cycles per instruction
ns	clock period in nano-seconds
wd sz adj	adjustment factor for word size: 32-bit = 1, 24-bit = 0.8, 16 bits: 0.67, 12 bits = 0.5, 8 bits = 0.33, 4 bits = 0.17, 1 bit = 0.04
std FOM	figure of merit used in massive open cores spreadsheet
TB	test bench present
TB OK	test bench passed
FPGA bd OK	runs on FPGA board
github location	web link to location of design archive
comments	